

Supply Chain & Inventory Analytics

Business Insights & Recommendations Report

Data Analyst Portfolio Project

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Management-focused interpretation of inventory, product, warehouse, supplier and procurement performance.

1. Executive Business Summary

The analysis provides a management view of supply chain performance across inventory, products, warehouses, suppliers, orders and procurement. The Power BI dashboard brings these areas together so decision-makers can identify operational risks and focus improvement efforts.

2. Major Business Insights

1. Inventory and Stockout Risk

Warehouses with elevated stockout activity require closer monitoring. Repeated stockouts can reduce sales opportunities and affect customer service. Inventory planning should therefore consider historical demand, current closing stock and replenishment requirements.

2. High-Demand Product Monitoring

Products with strong unit sales should receive higher inventory attention. Maintaining sufficient stock for high-demand products can reduce lost sales and improve service levels.

3. Supplier Fulfillment Performance

Supplier fulfillment rate is an important indicator of procurement reliability. Suppliers that repeatedly deliver less than the ordered quantity may create downstream inventory shortages.

4. Supplier Lead Time

Longer supplier lead times increase the amount of planning required before replenishment. Suppliers with consistently high lead times should be monitored and incorporated into reorder planning.

5. Late Delivery Risk

Late deliveries can affect production, inventory availability and customer fulfillment. Suppliers with higher late-delivery rates should be investigated for recurring operational issues.

6. Procurement Cost Control

Procurement spending should be monitored by supplier and warehouse. Comparing procurement expenditure across locations can help identify cost concentration and opportunities for better purchasing decisions.

7. Warehouse Performance

Warehouse-level analysis makes it possible to compare inventory, sales, stockouts and procurement activity. This supports targeted operational improvements rather than applying the same strategy to every warehouse.

8. Fulfillment Performance

Fulfillment metrics provide a direct view of operational execution. Lower fulfillment performance should trigger investigation into inventory availability, supplier delays, warehouse processes or order management.

3. Recommended Management Actions

- Prioritize warehouses with higher stockout rates for inventory review.
- Increase monitoring of high-demand products.
- Set supplier performance thresholds for fulfillment and late delivery.
- Review suppliers with consistently long lead times.
- Use supplier performance when making future procurement decisions.
- Analyze procurement costs by warehouse and supplier.

- Improve reorder planning using demand and lead-time information.
- Maintain safety stock for critical and high-demand products.
- Review recurring stockout and late-delivery patterns monthly.
- Use the Power BI dashboard as a recurring management-monitoring tool.

4. KPI Interpretation Guide

KPI	Management Interpretation
Stockout Rate	Higher values indicate greater inventory availability risk.
Inventory Turnover	Helps assess how efficiently inventory is moving.
Fulfillment Rate	Higher values generally indicate stronger order execution.
Procurement Fulfillment Rate	Shows how much of ordered quantity is actually received.
Average Lead Time	Indicates the typical supplier delivery duration.
Late Delivery Rate	Shows the proportion of procurement orders delivered later than expected.
Procurement Cost	Shows total purchasing expenditure and supports cost monitoring.

5. Overall Business Impact

The project creates a single analytical view of the supply chain. Instead of reviewing inventory, orders, suppliers and procurement separately, management can compare these areas together and identify relationships between demand, stock availability, supplier performance and procurement activity.

6. Conclusion

The analysis provides a practical foundation for improving inventory planning, supplier management, procurement efficiency and warehouse performance. The next analytical stage could include demand forecasting, ABC inventory classification, supplier risk scoring and Machine Learning-based demand prediction.

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